

Jhum check-point interpretation protocol

Mizoram jhum mapping — PlanetScope visual interpretation, 30 m check points. Keep this sheet open while labelling.

1. What counts as jhum for year Y

A plot is jhum for year Y if it was slashed and burned during the Y clearing season (about November Y–1 to May Y) and cropped in the Y monsoon.

- Judge ONLY what is inside the red 30 m square. Ignore the surroundings.
- More than half the square burned → 1. Half or less → 0.
- Clearing that began in November or January still belongs to year Y.
- A plot burned in Y–1 and merely cropped again in Y is NOT jhum for Y (label 0, last_year_jhum).

2. The four images to open, in order

Step	Planet monthly mosaic	A jhum plot for 2025 shows
1 Baseline	global_monthly_2024_10	Green: forest, bamboo or fallow, canopy intact
2 Slashing	global_monthly_2025_01	Cut vegetation drying: dull brown/grey, plot outline appears
3 Burning	global_monthly_2025_04	Black or dark brown surface, sharp edges, often on a slope
4 Cropping	global_monthly_2025_08	Patchy green crops, not closed canopy

If April is cloudy, check March and May before deciding. For the two-year question, also open global_monthly_2024_04.

3. Decision sequence (follow every time)

#	Question	If YES	If NO
1	Is the square bare/burned in the Oct baseline already?	0 — last_year_jhum, settlement, riverbed or landslide	go to 2
2	Does the square turn dark brown/black between Jan and May?	go to 3	0 — forest / bamboo / fallow
3	Are the edges sharp and the shape a discrete patch?	go to 4	0 — leaf_off (slope-wide, diffuse, no edge)
4	Is more than half the square inside that burn?	1 — jhum	0 — adjacent_jhum or partial_burn
5	Is it green again with patchy crops in Aug?	confirms 1	keep 1 but note no_crop_seen

4. The three mistakes that cost the most

Trap	Looks like	How to tell it apart
Dry-season leaf-off	Whole slopes turn pink/brown in Mar, green again by May	Diffuse, no sharp boundary, crown texture still visible, recovers within weeks. A burn stays dark and keeps hard edges
Adjacent jhum	A burn right beside the square; eye is drawn to it	Zoom so the red square fills the screen; judge only inside it
Second-year plot	Bare in Apr, crops in Aug — looks like jhum	Check the Oct baseline: a true new clearing is GREEN there, a second-year plot is already bare

5. What to type in each column

Column	Allowed values	Rule
ref_jhum	1 or 0 only	Never letters. 1 = jhum in the mapping year
ref_confidence	h, m, l	h = certain; m = probable; l = poor imagery or ambiguous. Never leave blank
ref_landcover	jhum, forest, bamboo, young_fallow, fallow, last_year_jhum, settlement, road, riverbed, landslide, wet_rice, plantation, bareland, water, other	Lower case, underscores, one value only. Must agree with ref_jhum: "jhum" only when ref_jhum = 1

Column	Allowed values	Rule
notes	cleared_YYYY_MM, cleared_two_years, edge_pixel, adjacent_jhum, partial_burn, leaf_off, forest_fire, cloud_in_MM, no_crop_seen	Use these fixed tags; add free text after them if needed

Consistency check before you finish: *ref_landcover = jhum* must always pair with *ref_jhum = 1*, and *ref_jhum = 1* must always pair with *ref_landcover = jhum*.

6. How to write the notes column

Notes record **why** you decided, not only what you saw. Write TAGS, not sentences: lower case, underscores, separated by semicolons, free text last if needed. Example: cleared_2025_01; edge_pixel; burn_covers_60pct

6.1 When it was cleared

Tag	Use it when
cleared_2024_10 ... cleared_2025_05	You can see the month the slashing or burning first appears
cleared_two_years	The plot was also bare or burned in the previous year's imagery
burn_between_mosaics	Green in one month, cropped in the next; no image caught the burn
late_burn	Still unburned in April, burned in May

6.2 How much of the 30 m square is affected

Tag	Use it when
burn_covers_90pct (also _75pct, _60pct, _50pct, _30pct)	Rough share of the square that is burned
edge_pixel	The square straddles the plot boundary
adjacent_jhum	Burn is nearby but outside the square
partial_burn	Burned patchily inside the square, not a clean clearing
two_plots_in_pixel	More than one small plot inside the square

6.3 Why you might be unsure

Tag	Use it when
cloud_in_04 (use the month number)	That month's mosaic is cloudy at this point
shadow_slope	Steep shaded slope, colours hard to judge
mosaic_seam	A basemap seam crosses the square
no_crop_seen	Burned, but August shows no clear crop growth
image_blur	The mosaic is soft or low quality here

6.4 What it is, if not jhum

Tag	Use it when
leaf_off	Deciduous browning: diffuse, slope-wide, recovers by May
bamboo_dry	Bamboo stand browning; fine, even texture
forest_fire	Burned, but inside continuous forest, no plot shape, no crops after
old_plot_regrowing	Old plot outline visible, vegetation recovering
second_year_crop	Cropped this year but cleared last year
permanent_field	Terraces or bunds, same shape every year
new_construction	Bare because of building or road cutting
river_sand	Bare sand along a channel
landslide_scar	Teardrop bare scar on a steep slope

6.5 Worked examples

Situation	Write this
Clear jhum, most of the square, slashed in January	cleared_2025_01; burn_covers_90pct
Burn only clips the corner of the square	edge_pixel; burn_covers_30pct
Burn beside the square; square itself is forest	adjacent_jhum; leaf_off
Bare already in October, cropped in August	second_year_crop; cleared_2024_12
April cloudy, March shows the burn	cloud_in_04; cleared_2025_03; burn_covers_75pct
Burned patch inside closed forest, no crops after	forest_fire; no_crop_seen
Steep shaded slope, probably burned	shadow_slope; burn_covers_50pct
Ordinary forest, nothing notable	leave blank

6.6 Rules and the minimum set

- Semicolons between tags, never commas (commas break CSV export).
- Two or three tags is plenty; fatigue causes worse errors than missing tags.
- Blank is fine for plain forest. Do not tag the obvious.
- If short on time, use only these three: **cleared_YYYY_MM** on every jhum point, **edge_pixel** when the square straddles a boundary, and **leaf_off** when brown vegetation is seasonal rather than burned. They give the burn-date distribution, the edge-error rate and the leaf-off confusion rate.

6.7 What each tag family yields for the paper

Tag family	Result it produces
cleared_YYYY_MM	Distribution of burn dates — evidence for the Nov–May window
cleared_two_years	How common consecutive cultivation is
edge_pixel, burn_covers_Npct	How much error comes from 30 m pixel size; what Sentinel-2 would fix
cloud_in_MM	How often the burn season is obscured
leaf_off, bamboo_dry	The main confusion classes, quantified
forest_fire	Whether the mapped class is jhum or burned area

7. Working rules that keep the validation valid

- Label each point once. Do not go back after learning what the map said.
- Do not open the KEY file, the index values, or the GEE inspector until all labelling is finished.
- Work in the given order; the points are already shuffled across strata.
- Never leave a point blank. If unsure, give your best judgement with confidence I.
- Save every 20 points (Ctrl+S). Toggle editing off before exporting.
- Take a break every ~50 points; fatigue is the main source of entry errors.

8. QGIS setup that prevents typing errors

Setting	Where	Effect
Value Map on ref_jhum (1 = Jhum, 0 = Not jhum)	Layer Properties → Attributes Form	Dropdown instead of typing
Value Map on ref_confidence and ref_landcover	same	No spelling variants
List widget on notes with the common tags	same	Pick tags instead of typing them
Read-only on point_id, year, lon, lat	same	Cannot be edited by accident
Categorized symbology on ref_jhum	Layer Properties → Symbology	Blank points stay visible until labelled
Scale locked at 1:4000–1:5000	Scale box, bottom bar	Same viewing scale for every point

9. Self-check before sending the file

- Count of labelled rows equals the number of points.

- ref_jhum contains only 1 and 0 (sort the column to spot stray letters).
- No row has ref_landcover = jhum with ref_jhum = 0, or the reverse.
- ref_confidence filled for every row.
- Re-interpret a random 10% a day later without looking at your first answers, and report the agreement rate.